

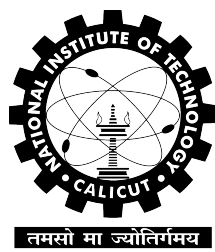
Metrics to Assess the Interpretable Speech Systems

CS4099 Project Final Report

Submitted by

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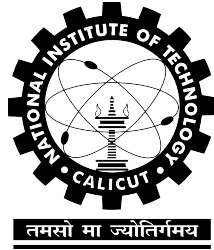


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April 15, 2025

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2025

CERTIFICATE

Certified that this is a bonafide record of the project work titled

Metrics to Assess the Interpretable Speech Systems

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*of eighth semester B. Tech in partial fulfillment of the requirements for the
award of the degree of Bachelor of Technology in Computer Science and
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DECLARATION

We hereby declare that the project titled, **Metrics to Assess the Interpretable Speech Systems**, is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or any other institute of higher learning, except where due acknowledgement and reference has been made in the text.

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Abstract

This work explores post hoc explanation methods' evaluation in voice systems to enhance model interpretability. To assist this study, we created a new Hindi speech-emotion dataset called BhavanaVani as part of the effort. Initially, baseline system performance was established using models for emotion detection and speech recognition. We thus generated binary masks over the speech input using LMAC (Local Model-Agnostic Classifier) and assessed these explanations using structural metrics and fidelity. High interpretability scores and high classification accuracy revealed by the results confirmed the value of our approach. Future additions comprise deep learning interpretability frameworks and multilingual datasets.

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Chapter 1

Introduction

Interpretable machine learning (IML) is a cornerstone in the development of modern artificial intelligence systems, particularly in critical domains where decision transparency is non-negotiable. In recent years, the advancement of deep learning has led to tremendous improvements in speech recognition and natural language processing. However, the complexity of these models has made it increasingly difficult to understand and explain their internal decision-making processes. This opacity can result in a lack of trust, poor user acceptance, and significant risks in high-stakes environments like health-care, law enforcement, and finance.

In speech-based systems, interpretability is crucial for diagnosing model errors, uncovering data biases, and ensuring fairness. Unlike structured data, audio signals are inherently temporal and noisy, which makes generating human-understandable explanations even more challenging. Additionally, speech emotion recognition and multilingual classification tasks demand systems that can not only perform accurately but also justify their predictions.

The primary goal of this project is to develop and assess a methodology that allows the evaluation of speech model explanations. We particularly focus on Local Model-Agnostic Classifier (LMAC), a post hoc explanation technique that generates binary masks to highlight which segments of an

audio input influence a model’s prediction.

These masks are analyzed using both faithfulness metrics which evaluate how accurately explanations reflect the model’s logic and structural metrics which assess the simplicity and alignment of the explanation with input structure.

Our work is guided by the need for interpretability in the context of deep learning-based speech systems. We build on the curated BhavanaVani dataset that contains labeled Hindi speech samples across multiple speakers and emotions. Through this approach, we aim to improve transparency in speech-based AI systems, making them more accountable and user-friendly. This study contributes to the ongoing efforts in XAI by providing a systematic framework to evaluate and optimize interpretable speech models.

Chapter 2

Literature Survey

Interpretability in machine learning is a well-established area of research, but its application to speech systems presents unique challenges. This chapter explores key works that have informed the development of this project.

2.1 Explainable AI (XAI)

The broader field of Explainable Artificial Intelligence (XAI) aims to bridge the gap between complex machine learning models and human understanding. As machine learning models—particularly deep neural networks—become more opaque, there is a growing need to make their decision-making processes transparent and interpretable, especially in domains involving high-stakes decisions.

One of the foundational definitions of interpretability was provided by Doshi-Velez and Kim (2017), who described it as "the ability to explain or to present in understandable terms to a human." Their work emphasized that interpretability is context-dependent and varies based on the end-user's goals—whether they are developers, regulators, or end-users.[\[1\]](#)

2.2 Local Interpretable Model-agnostic Explanations (LIME)

LIME, introduced by Ribeiro et al. (2016), is a post hoc explanation method that perturbs input features and trains a surrogate interpretable model locally around the prediction. While LIME has been widely adopted for structured data and images, its application to temporal domains like audio is less straightforward due to the continuous and high-dimensional nature of audio signals. Recent adaptations, such as EmoLIME, have tailored LIME for speech emotion recognition by identifying influential frequency ranges in audio data .[2]

2.3 SHAP (SHapley Additive exPlanations)

SHAP explains the output of any model using game theory and assigns importance values to each feature. While powerful, SHAP’s use in audio remains limited due to difficulty in segmenting audio into meaningful discrete features. Studies have applied SHAP in medical domains, such as Alzheimer’s disease detection, highlighting its potential in interpreting complex models .[3]

2.4 Grad-CAM and Saliency Maps for Audio

Visual-based techniques like Grad-CAM have been extended to audio spectrograms. These highlight important frequency-time regions that contribute to a model’s decision. However, they often lack resolution and may not generalize well across domains.[4]

2.5 LMAC for Speech

LMAC is a binary masking technique introduced specifically for audio data. It overlays masks on raw waveform or spectrogram inputs to identify segments relevant to a classifier’s prediction. Mask-in and Mask-out operations assess classifier confidence when explanations are preserved or removed, respectively. This technique offers more direct, intuitive insight into what parts of the audio the model “listens to.” [5]

2.6 Evaluation Metrics for Interpretability

Evaluating the quality of explanations in interpretable machine learning models, especially in domains like speech recognition and emotion detection, is crucial to ensuring that the explanations are both accurate and human-understandable. The evaluation metrics fall broadly into two categories:

- **Faithfulness Metrics:** Faithfulness metrics assess whether the explanation accurately reflects the model’s decision-making process.

L2I-Faithfulness: This metric measures the change in model prediction confidence when only the important (explained) parts of the input are retained.

$$\text{L2I-Faithfulness} = f(\mathbf{x}') - f(\mathbf{x})$$

Average drop: This metric evaluates how much the model’s confidence drops when we retain only the important parts of the input.

Average Gain: This measures how much the model confidence increases when irrelevant parts are removed.

- **Structural Metrics:** Structural metrics assess how simple, sparse, and interpretable the explanations are from a human point of view.

Sparseness: This refers to the fraction of input that has been marked as important.

Complexity: his measures the number of contiguous segments or regions marked as important.

Complexity = Number of contiguous non-zero segments in the mask

S-AUCN-AUC: Saliency AUC evaluates how well the explanation aligns with model confidence across saliency values.

S-AUC=Area Under ROC Curve(s(x),y)

Metric	Type	Interpretation
L2I-Faithfulness	Faithfulness	Measures confidence retention when only important input segments are retained.
Average Drop	Faithfulness	Measures how much the model's confidence decreases when unimportant parts are removed. Lower is better.
Average Gain	Faithfulness	Measures the increase in model confidence when irrelevant regions are removed. Higher is better.
Sparseness	Structural	Proportion of the input identified as important. Lower sparseness indicates more concise explanations.
Complexity	Structural	Number of contiguous segments marked as important. Lower values suggest easier human interpretation.
S-AUC (Saliency AUC)	Structural	Area under the ROC curve measuring alignment between saliency values and model confidence.
N-AUC (Normalized AUC)	Structural	S-AUC normalized against a random baseline. Higher values indicate better explanation quality.

Table 2.1: Summary of Evaluation Metrics for Interpretability

These metrics attempt to quantify how well the explanation reflects actual model behavior and how easily a human can make sense of it.[6]

2.7 Existing Datasets

Popular datasets such as RAVDESS, TESS, and CREMA-D have been extensively used for speech emotion recognition research. While these datasets have played a crucial role in advancing the field, they primarily consist of English-language speech with controlled studio conditions and limited speaker diversity. As a result, they do not sufficiently capture the temporal complexity, linguistic variation, or accentual diversity present in real-world speech—particularly in non-English contexts like Hindi.

To overcome these limitations, our work introduces BhavanaVani, a curated Hindi speech-emotion dataset designed specifically to address the gaps in linguistic and regional coverage. BhavanaVani includes samples from speakers with varied regional Hindi accents, thus enhancing the representativeness of the dataset. This linguistic diversity is essential for developing and evaluating Explainable AI (XAI) models in speech applications, especially in a multilingual country like India where dialectical variation is significant.

Moreover, BhavanaVani is annotated not just for emotion labels, but also for interpretability assessments, making it uniquely suited for analyzing and benchmarking post hoc explanation techniques such as LMAC. This enables a deeper evaluation of how well explanation methods align with both model predictions and human intuitions.

In summary, while many explanation techniques exist, few have been optimized for the temporal, linguistic, and subjective nature of speech data. Our work builds upon these techniques and adapts them through the lens of BhavanaVani, offering a rich, real-world foundation for interpretable speech modeling in Hindi.

This literature review demonstrates that while there are numerous explanation techniques, few are optimized for the temporal and subjective nature of speech. Our work builds on the strengths of these techniques and adapts them for use in spoken language applications with a strong focus on evaluation.

Chapter 3

Problem Definition

As speech systems become more complex and ubiquitous, the need to understand their decision-making logic grows exponentially. These systems are employed in real-time applications like virtual assistants, customer service bots, emotion recognition platforms, and medical voice diagnostics. While deep learning-based models yield high accuracy, their black-box nature leads to reduced transparency, poor accountability, and limited user trust.

The problem addressed in this project is the absence of a standardized mechanism to measure and validate the interpretability of explanations generated by speech models. Although post hoc techniques like LIME[2] and LMAC[3] exist, their adaptation to audio remains insufficient.

The project seeks to address the following research questions:

1. How can we determine the faithfulness of an explanation generated for a speech model?
2. What structural qualities (e.g., sparseness, alignment with perceptual boundaries) define a good explanation in audio?

To address the lack of standardization in evaluating interpretability in speech models, this project introduces a systematic framework based on faithfulness and structural metrics. By first deploying speech recognition

and emotion detection models and then applying the LMAC (Local Model-Agnostic Classifier) technique, we generate time-aligned binary masks that highlight the most influential parts of the audio input. These masks are quantitatively evaluated using metrics like L2I-Faithfulness, Average Drop/Gain, Sparseness, and S-AUC, providing both objective and perceptual insight into explanation quality. The creation of the BhavanaVani dataset, which includes regionally accented Hindi speech and interpretability annotations, further enhances our ability to benchmark and analyze explanations in a linguistically rich and underexplored domain. This comprehensive approach enables deeper understanding and trust in speech-based AI systems.

Chapter 4

Methodology

4.1 Dataset Source

In this project, two datasets were utilized for developing and evaluating the emotion recognition model. One is a custom dataset specifically created for this work — the **BhavanaVani** dataset. The other is a publicly available dataset — the **Speech Emotion Recognition Hindi** dataset from Kaggle. These datasets collectively contributed to the robustness and generalizability of the proposed model.

4.1.1 BhavanaVani: Custom Hindi Speech Emotion Dataset

The "BhavanaVani" database contains speech recordings from five different regions of India: NORTH, SOUTH, EAST, WEST, and CENTRAL. These regions include both native and non-native Hindi speakers.

"BhavanaVani" was collected by reaching out to a few students and staff members from NIT Calicut, as well as friends from other colleges who speak Hindi. The sentences in this dataset are relatively short, which makes them suitable for capturing and displaying accents at the sentence level.

Region	Number of Samples	Duration
North	11	5 min 5 sec
South	8	2 min 1 sec
East	10	5 min 9 sec
West	9	3 min 2 sec
Central	9	7 min 5 sec
Total	47	22 min

Table 4.1: Region-wise Distribution of Samples in BhavanaVani Dataset

While the BhavanaVani dataset was meticulously curated to capture a wide range of emotions in Hindi speech, it inherently suffers from potential biases due to its limited speaker pool, which results in repeated and biased emotional expressions. To address this limitation and enhance the robustness of the model, the publicly available *Speech Emotion Recognition Hindi* dataset from Kaggle was incorporated. This additional dataset introduces greater speaker diversity, environmental variability, and real-world noise factors, thereby enabling the model to learn more generalized emotional patterns and reducing the risk of dataset-specific overfitting.

4.1.2 Speech Emotion Recognition Hindi: Kaggle Dataset

Overview

The **Speech Emotion Recognition Hindi** dataset, curated by Vishal Bhardwaj and hosted on Kaggle, is designed to facilitate the development and evaluation of models for recognizing emotions in Hindi speech. This dataset serves as a valuable resource for researchers and practitioners aiming to advance speech emotion recognition (SER) in the context of Hindi language audio data.^[7]

Table 4.2: Participant Details in Kaggle Dataset

S.No	Name	Height (in feet)	Weight (kg)	Age	Gender
1	Deepika Rajwade	5.4"	60kg	23	F
2	Nidhi Avasthi	5.2"	54kg	22	F
3	Sonam Sahu	5"	50kg	22	F
4	Kajal Chandra	5"	55kg	22	F
5	Som Sahu	5.6"	60kg	23	M
6	Jayprakash Jangde	6"	100kg	23	M
7	Ramendra Bhargav	5.11"	68kg	23	M
8	Karan Patel	5.6"	65kg	24	M

Dataset Structure

The dataset comprises audio recordings in Hindi, systematically organized to represent various emotional states. Each audio file (.wav) is labeled with the corresponding emotion it conveys. The dataset is structured into multiple directories, each potentially representing a specific emotion category. This organization aids in efficient data handling and model training processes.

Folder Hierarchy

Dataset Structure: The dataset follows a hierarchical organization structured around three primary levels: **actors**, **sessions**, and **emotions**. At the top level, each speaker is identified by a unique **actor ID** (e.g., *Actor 1*, *Actor 2*, etc.), which helps in tracking speaker-specific variations in emotional expression. Within each actor's directory, there are multiple **session folders** (e.g., *Session 1* to *Session 10*), representing different recording instances. These sessions capture the full set of emotion-labeled sentences across varied temporal contexts or repetitions. Inside each session folder, the data is further divided into subfolders named after individual **emotions** (such as *anger*, *happy*, *sad*, etc.), and each of these folders contains the corre-

sponding audio files exhibiting the specified emotional state. This structured layout facilitates systematic processing, emotion-wise access, and speaker-based analysis, making the dataset suitable for both supervised learning and speaker-dependent studies in emotion recognition.

Naming Convention

File Naming Convention and Storage Format: Each audio file in the dataset follows a consistent and descriptive naming pattern to facilitate easy identification and parsing. The naming format is defined as: `actorID.sessionID.emotion-sentenceNumber.wav`. For example, the file `1.1.anger-01.wav` represents a recording from *Actor 1*, recorded during *Session 1*, expressing the emotion *anger*, and corresponding to *Sentence 1* in the dataset. This systematic naming convention enables automated extraction of metadata such as speaker identity, session context, and emotion class directly from the filename, aiding in preprocessing and data organization.

All audio files are stored in the standard **WAV** format to ensure high-quality, lossless audio preservation. The recordings are in **Hindi**, the target language for the emotion recognition task. Each file is recorded in **mono channel** format with a **sampling rate of 16kHz**, which is widely adopted in speech processing applications. The **bit depth** of each file is **16-bit**, offering a good balance between audio fidelity and storage efficiency. This consistent format ensures compatibility with most audio processing libraries and maintains the integrity of the emotional cues embedded in the speech.

Data Collection Guidelines

Actors were instructed to speak all 10 sentences in each emotion clearly and consistently. Sessions were repeated over 10 instances to capture variation while maintaining naturalness. Recordings were stored in emotion-specific folders within each session directory.

Emotional Categories

Emotional Categories: Although the dataset’s official documentation does not explicitly list the exact number and nature of emotional categories, it aligns with the conventions followed by most Speech Emotion Recognition (SER) datasets. Typically, such datasets encompass a diverse set of fundamental emotions that cover a wide spectrum of human affective states. These commonly include: *anger*, *happiness*, *sadness*, *neutral*, *fear*, *disgust*, and *surprise*. The inclusion of these distinct categories ensures that the dataset supports a broad and nuanced understanding of emotional expression in speech. By covering both high-arousal (e.g., anger, happiness) and low-arousal (e.g., sadness, neutral) states, along with emotions that vary in valence, the dataset facilitates the training of models capable of capturing subtle acoustic and prosodic variations. This makes it well-suited for real-world applications such as human-computer interaction, affective computing, and conversational AI systems.

Data Format and Accessibility

The audio files are likely provided in standard formats such as WAV or MP3, ensuring compatibility with common audio processing tools. The dataset is publicly accessible via Kaggle and is intended for educational and research purposes. Users are encouraged to review the dataset’s license and usage terms on its Kaggle page: <https://www.kaggle.com/datasets/vishlb/speech-emotion-recognition-hindi>.

Applications

This dataset plays a crucial role in training and evaluating machine learning models across various applications. It is particularly valuable for Speech Emotion Recognition, enabling the development of algorithms that can accurately classify emotions based on speech signals. Additionally, it supports

advancements in Human-Computer Interaction by helping systems better understand and respond to users' emotional states. The dataset also contributes to language processing research, offering insights into emotional expression in Hindi speech and facilitating deeper linguistic analysis.

Limitations and Considerations

The dataset, while valuable, presents certain limitations that researchers should consider during model development and analysis. One key concern is the variation in the number of samples across emotion categories, which may lead to imbalanced training and affect model performance. Additionally, inconsistencies in recording conditions—such as differences in audio quality and background noise—could introduce variability in the data. Another limitation is the lack of detailed information regarding speaker diversity, including age, gender, and regional dialects, which may restrict the generalizability of models trained on this dataset. Addressing these challenges is essential for developing robust and accurate emotion recognition systems.

4.2 Audio Preprocessing

Audio files are resampled to 16kHz using Librosa, with silence trimming and volume normalization. We also segment the audio into consistent durations to ensure uniform feature extraction.

Cleaning and post-processing audio data is essential to improve the quality and usability of a dataset, particularly for applications like speech recognition. Below are the steps involved along with the tools used:

1. **Noise Removal:** Removing unwanted background noise is essential to make the speech samples clearer. This involves identifying and filtering out noises like hums, clicks, and environmental sounds (e.g., traffic or chatter) from the recordings.

2. **Organization by Accent:** After cleaning the data, the next task is to organize the recordings according to the speaker’s accent. This helps in training models that can recognize and differentiate various regional accents. In datasets like *BhavanaVani*, this step involves categorizing the samples into distinct groups based on the region (e.g., **NORTH, SOUTH, EAST, WEST, CENTRAL**).

4.3 Feature Extraction

We extract Mel-Frequency Cepstral Coefficients (MFCCs), which capture the timbral texture of speech. Each file produces 13 MFCCs averaged across time, yielding a fixed-size feature vector per sample. PCA is optionally applied for dimensionality reduction.

4.3.1 Objective

The goal of this stage is to convert raw audio signals into a structured numerical representation that can be effectively used for training a machine learning model. Mel Frequency Cepstral Coefficients (MFCCs) are extracted from the speech samples to capture the essential characteristics of the audio signal, which are then used as features for emotion classification.

4.3.2 Data and Directory Structure

The cleaned audio files are organized within the directory located at `/Desktop/emotion project/cleaned`, providing a centralized and accessible storage path for processed data. Each audio file is associated with its corresponding emotion label through the `emotions.csv` file, which serves as a structured mapping reference. This setup follows a one-to-one correspondence between audio filenames and their respective emotion labels, ensuring clarity and consistency in data annotation for further processing and model training.

4.3.3 MFCC Feature Extraction Process

The feature extraction process was implemented using the Librosa and NumPy libraries to convert raw audio data into meaningful input for classification models. For each audio file listed in `emotions.csv`, the full file path was constructed, and the audio was loaded using `librosa.load()` with a target sampling rate of 16 kHz—a standard in speech processing that ensures consistency across samples. Mel-Frequency Cepstral Coefficients (MFCCs) were then extracted using `librosa.feature.mfcc(y=y, sr=sr, n_mfcc = 13)`.

4.3.4 Feature Dataset Construction

Once all audio files have been successfully processed, the extracted MFCC feature vectors are compiled into a structured `pandas.DataFrame`, facilitating easy data manipulation and analysis. To associate each feature vector with its corresponding emotion category, a new column labeled `label` is appended to the `DataFrame`. This column contains the emotion class for each sample, as specified in the original `emotions.csv` file. The resulting `DataFrame`, which combines the numerical MFCC features with their corresponding emotion labels, represents the complete preprocessed dataset ready for model training. To preserve this data for future use, the `DataFrame` is saved as a CSV file.

4.4 Model Training

4.4.1 Objective

The primary objective of the model training phase is to construct a robust, generalizable deep learning model for real-time emotion classification from speech. The model should not only achieve high classification accuracy but also adapt to the nuances of audio-based emotion features. A hybrid architecture using Convolutional Neural Networks (CNN), Gated Recurrent Units

(GRU), and an attention mechanism is developed to capture both spatial and sequential patterns in the extracted acoustic features.

4.4.2 Feature Preparation and Dataset Loading

Each audio sample is transformed into a comprehensive 374-dimensional feature vector by extracting a diverse set of acoustic features, including MFCCs, Mel-spectrogram, Chroma features, Spectral Contrast, Tonnetz, pitch, and energy-based statistics. These features collectively capture both the spectral and temporal dynamics of the speech signal, making them highly effective for emotion classification tasks. To ensure that the features are on a consistent scale and improve the convergence of learning algorithms, the entire feature set is standardized using `StandardScaler()`, which normalizes the data to have zero mean and unit variance. Emotion labels corresponding to each audio sample are then encoded using `LabelEncoder()`, converting categorical class labels into numeric values suitable for model training. Finally, the preprocessed dataset is split into training and validation subsets using an 80-20 random split. This division allows the model to learn from the majority of the data while reserving a portion for unbiased performance evaluation during validation.

4.4.3 Model Architecture

The proposed model employs a CNN-GRU-Attention architecture designed to effectively capture both local and long-range patterns in speech signals for emotion recognition. The architecture begins with a Convolutional Neural Network (CNN) block, which consists of two 1D convolutional layers. Each convolutional layer is followed by Batch Normalization, ReLU activation, and Adaptive Average Pooling. This block is responsible for capturing short-term spectral features and local variations in the audio signal. Following the CNN layers, a bi-directional Gated Recurrent Unit (GRU) block with two layers

is used to model long-range dependencies and temporal dynamics across the entire audio sequence. To enhance the model’s focus on emotionally relevant segments of the input, a learnable attention mechanism is applied over the GRU outputs. This attention layer assigns dynamic weights to different time steps, allowing the network to selectively concentrate on the most emotionally salient parts of the speech. The final classification is performed by a fully connected head composed of two dense layers with dropout regularization, which mitigates overfitting. The model ultimately outputs a probability distribution over eight emotion classes, making it well-suited for nuanced speech emotion recognition tasks.

4.4.4 Training Setup

The model training process incorporates several techniques to ensure robust and efficient learning, especially in the presence of class imbalance. A weighted cross-entropy loss function is employed, where the class weights are set inversely proportional to their frequencies in the dataset. This approach ensures that minority classes are not underrepresented during training, thereby improving the model’s ability to generalize across all emotion categories. Optimization is carried out using the Adam optimizer with an initial learning rate of 0.001, which provides adaptive learning capabilities and accelerates convergence. To further refine the training dynamics, a StepLR learning rate scheduler is applied, reducing the learning rate by a factor of 0.5 every 15 epochs. This scheduled decay helps the model fine-tune its parameters as training progresses. The model is trained for a total of 60 epochs using a mini-batch size of 32, allowing efficient gradient updates while maintaining training stability. Wherever possible, GPU acceleration is utilized to significantly speed up the training process, enabling faster experimentation and improved computational efficiency.

4.4.5 Validation and Performance Monitoring

After each training epoch, the model undergoes a thorough evaluation on the validation set to assess its performance and guide further optimization. Several key metrics are computed during this process. Overall accuracy is measured as the ratio of correctly predicted samples to the total number of predictions, providing a general sense of the model’s effectiveness. In parallel, loss values for both the training and validation sets are recorded and plotted across epochs to monitor the model’s convergence behavior and detect signs of overfitting or underfitting. To gain a deeper understanding of how well the model distinguishes between different emotions, class-wise precision is calculated, revealing the model’s discriminative power for each individual emotion class. Additionally, a confusion matrix is generated and visualized, highlighting specific areas of misclassification by comparing predicted versus actual labels. This diagnostic tool is particularly useful for identifying patterns in the model’s errors, guiding targeted improvements in model design or dataset preprocessing.

4.4.6 Model Persistence

The best-performing model (based on validation accuracy) is saved as `best_model.pt`. The associated label encoder and feature scaler are saved using `joblib` to ensure consistent pre-processing during inference.

4.5 Local Model-Agnostic Classification (LMAC)

Local Model-Agnostic Classification (LMAC) is an interpretability framework used to explain decisions made by a trained machine learning model in a post-hoc and model-agnostic manner. In this project, we implement a variant of LMAC adapted to audio-based emotion recognition, where the goal is to identify the most influential regions of a speech signal responsible

for the predicted emotion.[\[5\]](#)

4.5.1 Motivation

In deep learning models such as CNNs or GRUs, the internal feature representations are often opaque. LMAC helps bring interpretability by identifying which parts of the input audio signal had the highest contribution to the model’s final prediction. This is essential for both trust and transparency in applications like emotion-aware voice interfaces, digital mental health tools, and human-computer interaction.

4.5.2 Working Mechanism

Our implementation of Local Model-Agnostic Classification (LMAC) provides interpretability by analyzing how different segments of the raw audio signal influence the model’s prediction. A fixed-size window is then systematically slid across the audio waveform, and in each step, a small segment of the signal is masked out (i.e., locally zeroed). For each masked segment, the model’s prediction is recalculated to measure the drop in confidence for the originally predicted class. This drop is interpreted as an importance score, quantifying how critical that specific segment was to the model’s decision. By aggregating these scores across all segments, LMAC produces a saliency-like importance map over time, highlighting which parts of the audio were most influential in the classification. This interpretability technique offers valuable insights into the model’s behavior and helps validate that predictions are driven by emotionally relevant regions of the speech signal.

4.5.3 Masking Strategies

To enhance interpretability through LMAC, two complementary optimization strategies are employed: Mask-Out and Mask-In. In the Mask-Out

approach, each segment of the audio—defined by a sliding window—is individually removed (zeroed out), and the resulting change in prediction confidence for the originally predicted class is measured. A significant drop in confidence suggests that the removed segment played a crucial role in the classification decision, thereby indicating high importance. Conversely, the Mask-In strategy, which is optional, retains only the current windowed segment while masking out the remainder of the audio. If the model is still able to maintain a high confidence in the original prediction based solely on that segment, it implies that the segment contains the core discriminative features necessary for classification. Together, these two strategies provide a robust mechanism for identifying and verifying emotionally salient regions in the speech signal, offering deeper insight into the model’s decision-making process.

4.5.4 Implementation Details

In our LMAC-based interpretability framework, we apply a **sliding window** technique over each 10-second audio input using a **window size of 0.2 seconds** and a **stride of 0.1 seconds**. This configuration ensures that each segment of the audio is systematically evaluated for its contribution to the model’s prediction. For each windowed segment t , an **importance score** I_t is calculated using the following equation:

$$I_t = P_{\text{base}}^{(c)} - P_{\text{masked}}^{(c)}$$

where $P_{\text{base}}^{(c)}$ denotes the original confidence score for the predicted class c when the full audio is used, and $P_{\text{masked}}^{(c)}$ is the confidence score when the segment t is masked. This difference quantifies the impact of removing the segment on the model’s decision, with larger values indicating higher importance. After computing these scores across all segments, the values are **normalized** to allow consistent interpretation. The final importance scores

are then visualized as a **time-aligned bar graph** superimposed over the waveform, highlighting the most influential regions in the audio signal that contributed to the emotion prediction.

4.5.5 Visualization of LMAC

The output of the LMAC analysis is displayed in the Streamlit app as a bar graph, aligned with the audio timeline. This allows users to clearly see which parts of their speech (e.g., a stressed word or vocal inflection) were most influential in the model’s decision.

4.5.6 LMAC Architecture

The architecture used in our model training remains unchanged. LMAC is applied post-hoc and does not modify the model. Instead, it works by querying the model multiple times with locally masked inputs. The entire flow is illustrated in Figure 4.1.

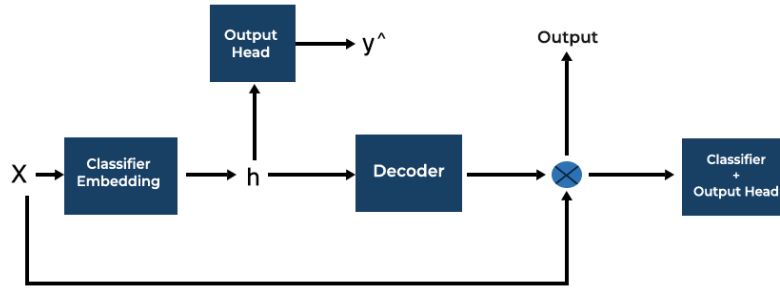


Figure 4.1: Architecture Flow for LMAC: From input audio through masking and prediction difference-based importance estimation

4.5.7 Benefits and Limitations

One of the key strengths of the Local Model-Agnostic Classification (LMAC) technique is its **model-agnostic** nature—it can be applied to any classifier regardless of its internal architecture, making it highly flexible. Additionally, LMAC provides **local explanations** by identifying specific time regions within the audio signal that are critical for the model’s prediction. This granularity helps in understanding which parts of the input contribute most significantly to the output. Furthermore, the method supports **fast inference** in real-time settings, making it suitable for applications involving live audio streams or interactive systems.

However, LMAC is not without limitations. One drawback is the **computational overhead** involved, especially when processing long-duration audio files, as it requires multiple forward passes through the model for each masked segment. Moreover, the accuracy and reliability of the importance scores may be influenced by **audio noise or segment misalignments**, which can distort the true contribution of certain regions in the audio. These factors should be considered when deploying LMAC in practical or noisy environments.

4.6 Evaluation Metrics

In the context of speech and emotion detection, interpretability metrics help assess whether the model is attending to the correct parts of the input signal—such as prosodic cues, pitch, or meaningful acoustic regions—when making a prediction.

These metrics also evaluate how explanations align with human intuition, linguistic structure, and known relevant features in speech signals.

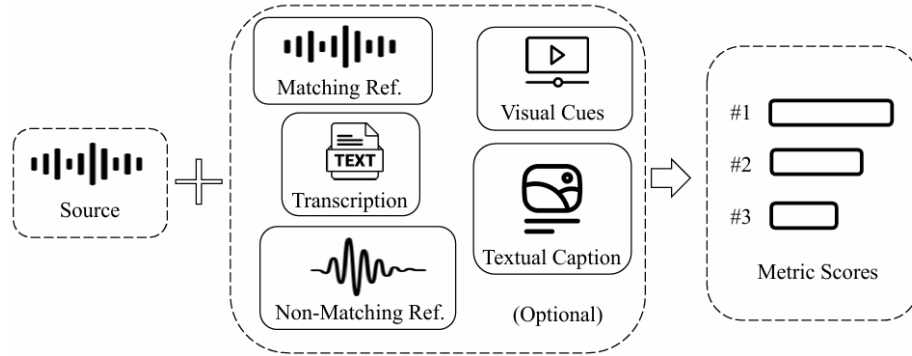


Figure 4.2: Metric Evaluation Pipeline: The source audio is evaluated using various reference signals including transcription, visual cues, textual captions (optional), and reference audio (matching or non-matching), producing final metric scores.

This report incorporates various evaluation metrics such as L2I-Faithfulness, Average Gain/Drop, Structure-AUC (S-AUC), Noise Retention, and Semantic Alignment. Each of these metrics serves a specific purpose in evaluating the fidelity, relevance, and clarity of model explanations. For example, L2I-Faithfulness measures the sensitivity of model predictions to changes in the highlighted explanation region, while S-AUC quantifies how well the explanation aligns with known speech-relevant areas.

Table 4.3: Evaluation Metrics for Explanation Assessment

Metric	Type	Purpose	Description
L2I-Faithfulness	Quantitative	Measures how faithful the explanation is to the model’s prediction	Evaluates the change in model prediction when parts of the input highlighted by explanations are perturbed or removed.
Average Gain / Drop	Quantitative	Analyzes the performance variation due to explanation masking	Computes how much the model’s confidence or accuracy changes when top salient regions are masked-in or masked-out.
S-AUC (Structure-AUC)	Quantitative	Measures structural relevance of explanations	Computes Area Under Curve (AUC) to quantify overlap between highlighted regions and ground-truth speech-relevant segments.
Noise Retention	Quantitative	Identifies if irrelevant (background) noise is being included in the explanation	Measures how much of the background (non-speech) region is retained in the explanation mask. Lower is better.
Explanation Sparsity	Quantitative	Indicates how focused or distributed the explanation is	Measures the number of features or regions used to form the explanation. Useful for evaluating interpretability.
Semantic Alignment	Qualitative	Measures alignment of explanation with human reasoning	Subjective check for whether explanations align with human-perceived important features (e.g., prosody, emotion).

Chapter 5

Results

This chapter presents the results obtained after training and evaluating our CNN-GRU-Attention based Speech Emotion Recognition system. The evaluation covers both the performance of the classifier and the interpretability of the model using Local Model-Agnostic Classification (LMAC). The training was carried out on extracted acoustic features, including MFCCs, Chroma, Mel Spectrogram, Tonnetz, Contrast, Pitch, and Energy features, yielding a 374-dimensional input per sample.

5.1 Training Progress

The model was trained for 60 epochs using the Adam optimizer and categorical cross-entropy loss function. The following metrics and visualizations illustrate the model's learning curve:

- **Training Accuracy:** Peaked at **99.71%**.
- **Validation Accuracy:** Reached **72.34%**.
- **Training Loss:** Decreased steadily over epochs, showing convergence.

```

saurav@saurav-HP:~/Desktop/finalyearproject/scripts$ python3 train_model.py
Loaded data: (6400, 374), labels: 6400
Epoch 1/60 - Train Acc: 0.1695, Val Acc: 0.2539
✓ Best model saved!
Epoch 2/60 - Train Acc: 0.2477, Val Acc: 0.2891
✓ Best model saved!
Epoch 3/60 - Train Acc: 0.2746, Val Acc: 0.3063
✓ Best model saved!
Epoch 4/60 - Train Acc: 0.3168, Val Acc: 0.3180
✓ Best model saved!
Epoch 5/60 - Train Acc: 0.3521, Val Acc: 0.3508
✓ Best model saved!
Epoch 6/60 - Train Acc: 0.3883, Val Acc: 0.3906
✓ Best model saved!
Epoch 7/60 - Train Acc: 0.4264, Val Acc: 0.4266
✓ Best model saved!
Epoch 8/60 - Train Acc: 0.4590, Val Acc: 0.4578
✓ Best model saved!
Epoch 9/60 - Train Acc: 0.5016, Val Acc: 0.4617
✓ Best model saved!
Epoch 10/60 - Train Acc: 0.5340, Val Acc: 0.5102
✓ Best model saved!
Epoch 11/60 - Train Acc: 0.5787, Val Acc: 0.5148
✓ Best model saved!
Epoch 12/60 - Train Acc: 0.6240, Val Acc: 0.5445
✓ Best model saved!
Epoch 13/60 - Train Acc: 0.6592, Val Acc: 0.5383
Epoch 14/60 - Train Acc: 0.6865, Val Acc: 0.5781
✓ Best model saved!
Epoch 15/60 - Train Acc: 0.7357, Val Acc: 0.5938
✓ Best model saved!
Epoch 16/60 - Train Acc: 0.8045, Val Acc: 0.6312
✓ Best model saved!
Epoch 17/60 - Train Acc: 0.8484, Val Acc: 0.6602
✓ Best model saved!
Epoch 18/60 - Train Acc: 0.8652, Val Acc: 0.6766
✓ Best model saved!
Epoch 19/60 - Train Acc: 0.8934, Val Acc: 0.6617
Epoch 20/60 - Train Acc: 0.9059, Val Acc: 0.6828
✓ Best model saved!
Epoch 21/60 - Train Acc: 0.9252, Val Acc: 0.6797
Epoch 22/60 - Train Acc: 0.9314, Val Acc: 0.6773
Epoch 23/60 - Train Acc: 0.9354, Val Acc: 0.6820
Epoch 24/60 - Train Acc: 0.9533, Val Acc: 0.6969
✓ Best model saved!
Epoch 25/60 - Train Acc: 0.9563, Val Acc: 0.6977
✓ Best model saved!
Epoch 26/60 - Train Acc: 0.9625, Val Acc: 0.7094
✓ Best model saved!
Epoch 27/60 - Train Acc: 0.9672, Val Acc: 0.7000
Epoch 28/60 - Train Acc: 0.9742, Val Acc: 0.7039
Epoch 29/60 - Train Acc: 0.9787, Val Acc: 0.7047
Epoch 30/60 - Train Acc: 0.9795, Val Acc: 0.6922

```

Figure 5.1: Training and Validation Accuracy per Epoch

```

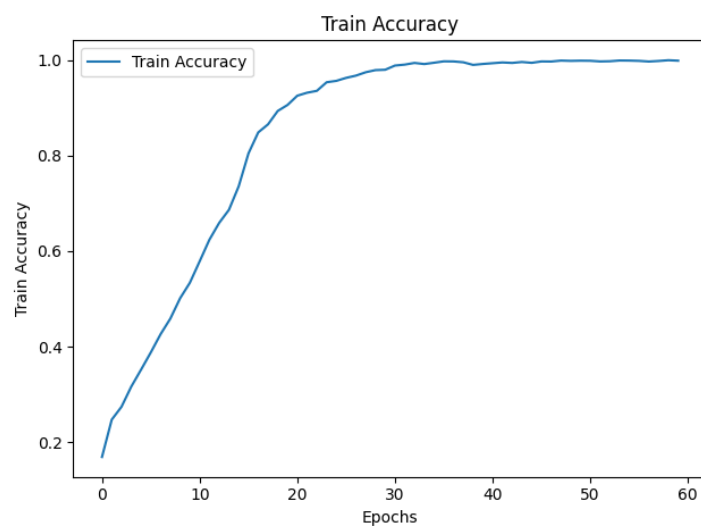
Epoch 31/60 - Train Acc: 0.9883, Val Acc: 0.7102
✓ Best model saved!
Epoch 32/60 - Train Acc: 0.9902, Val Acc: 0.7188
✓ Best model saved!
Epoch 33/60 - Train Acc: 0.9938, Val Acc: 0.7188
Epoch 34/60 - Train Acc: 0.9914, Val Acc: 0.7180
Epoch 35/60 - Train Acc: 0.9941, Val Acc: 0.7086
Epoch 36/60 - Train Acc: 0.9971, Val Acc: 0.7234
✓ Best model saved!
Epoch 37/60 - Train Acc: 0.9969, Val Acc: 0.7125
Epoch 38/60 - Train Acc: 0.9951, Val Acc: 0.7070
Epoch 39/60 - Train Acc: 0.9896, Val Acc: 0.7094
Epoch 40/60 - Train Acc: 0.9916, Val Acc: 0.7047
Epoch 41/60 - Train Acc: 0.9932, Val Acc: 0.7148
Epoch 42/60 - Train Acc: 0.9947, Val Acc: 0.7047
Epoch 43/60 - Train Acc: 0.9938, Val Acc: 0.7031
Epoch 44/60 - Train Acc: 0.9957, Val Acc: 0.7102
Epoch 45/60 - Train Acc: 0.9939, Val Acc: 0.7094
Epoch 46/60 - Train Acc: 0.9969, Val Acc: 0.7211
Epoch 47/60 - Train Acc: 0.9967, Val Acc: 0.7102
Epoch 48/60 - Train Acc: 0.9986, Val Acc: 0.7141
Epoch 49/60 - Train Acc: 0.9980, Val Acc: 0.7148
Epoch 50/60 - Train Acc: 0.9984, Val Acc: 0.7094
Epoch 51/60 - Train Acc: 0.9982, Val Acc: 0.7109
Epoch 52/60 - Train Acc: 0.9969, Val Acc: 0.7063
Epoch 53/60 - Train Acc: 0.9973, Val Acc: 0.7109
Epoch 54/60 - Train Acc: 0.9988, Val Acc: 0.7109
Epoch 55/60 - Train Acc: 0.9986, Val Acc: 0.7164
Epoch 56/60 - Train Acc: 0.9980, Val Acc: 0.7180
Epoch 57/60 - Train Acc: 0.9967, Val Acc: 0.7172
Epoch 58/60 - Train Acc: 0.9979, Val Acc: 0.7133
Epoch 59/60 - Train Acc: 0.9994, Val Acc: 0.7227
Epoch 60/60 - Train Acc: 0.9984, Val Acc: 0.7164
✓ Training complete. All graphs and model files saved

```

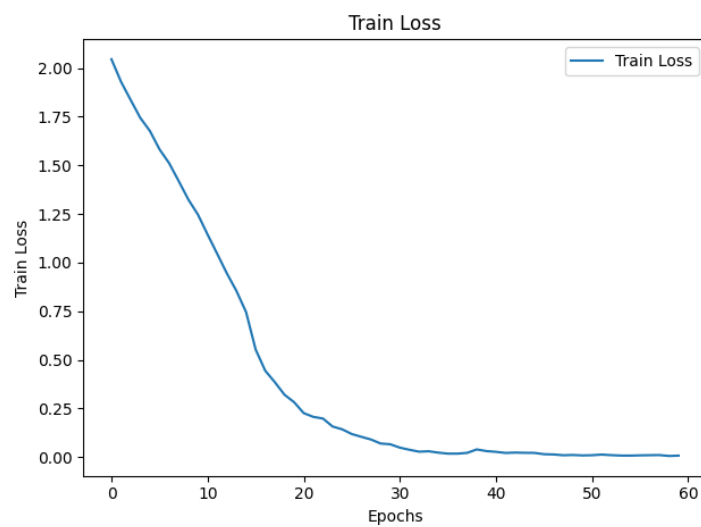
Figure 5.2: Training and Validation Accuracy per Epoch

To evaluate the training dynamics of the proposed model, the training and validation accuracy and loss curves over 60 epochs are presented in Figures 5.3 and 5.4. The training accuracy steadily increased and reached nearly 100% by epoch 35, while the corresponding training loss decreased smoothly, indicating effective learning and convergence.

However, the validation accuracy plateaued around 72%, and the validation loss began to increase after epoch 20, suggesting the onset of overfitting. This divergence between training and validation metrics highlights the model's strong learning capability on the training set but also reveals a gap in generalization to unseen data. These insights emphasize the need for further regularization, data augmentation, or early stopping to mitigate overfitting and improve model robustness.

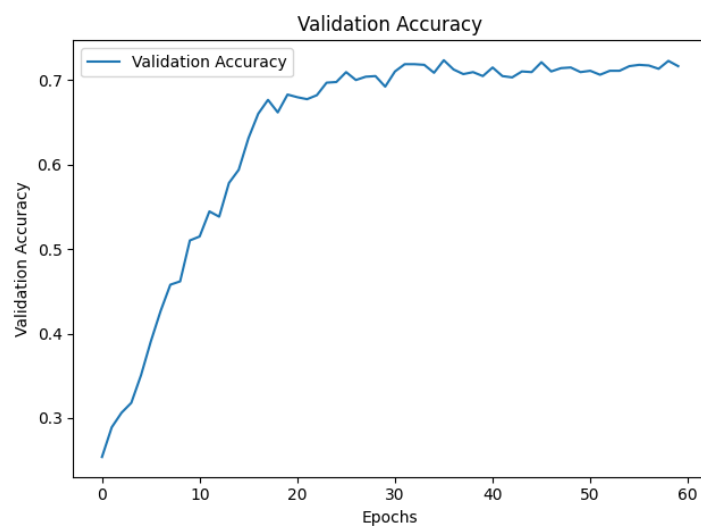


(a) Training Accuracy

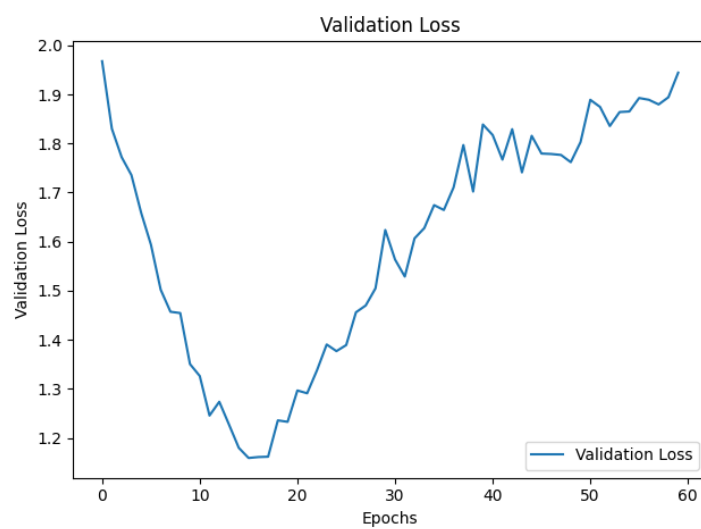


(b) Training Loss

Figure 5.3: Training accuracy and loss over epochs



(a) Validation Accuracy



(b) Validation Loss

Figure 5.4: Validation accuracy and loss over epochs

5.2 Test Set Evaluation

The model was evaluated on the ‘test_{wavs}/‘datasetcontaining100unseenaudiofilesspanningeighte

The confusion matrix is a tabular representation that is used to evaluate the performance of classification models. It includes four key components: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). The values on the diagonal of the matrix, TP and TN, indicate the correct predictions for each class. In contrast, the off-diagonal elements, FP and FN, correspond to incorrect classifications. Based on these values, various evaluation metrics are calculated, as detailed below.

$$\text{Accuracy} = \frac{\text{True Positive (TP)} + \text{True Negative (TN)}}{\text{Total Positive (P)} + \text{Total Negative (N)}} \quad (5.1)$$

$$\text{Recall (Sensitivity)} = \frac{\text{True Positive (TP)}}{\text{True Positive (TP)} + \text{False Negative (FN)}} \quad (5.2)$$

$$\text{Precision} = \frac{\text{True Positive (TP)}}{\text{True Positive (TP)} + \text{False Positive (FP)}} \quad (5.3)$$

$$F_1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5.5)$$

The results of the classification are visualized using a confusion matrix. In the matrix, the rows represent the actual classes and the columns represent the predicted classes. These metrics allow us to evaluate not just the accuracy, but also the reliability and robustness of our classifier across all emotion categories.

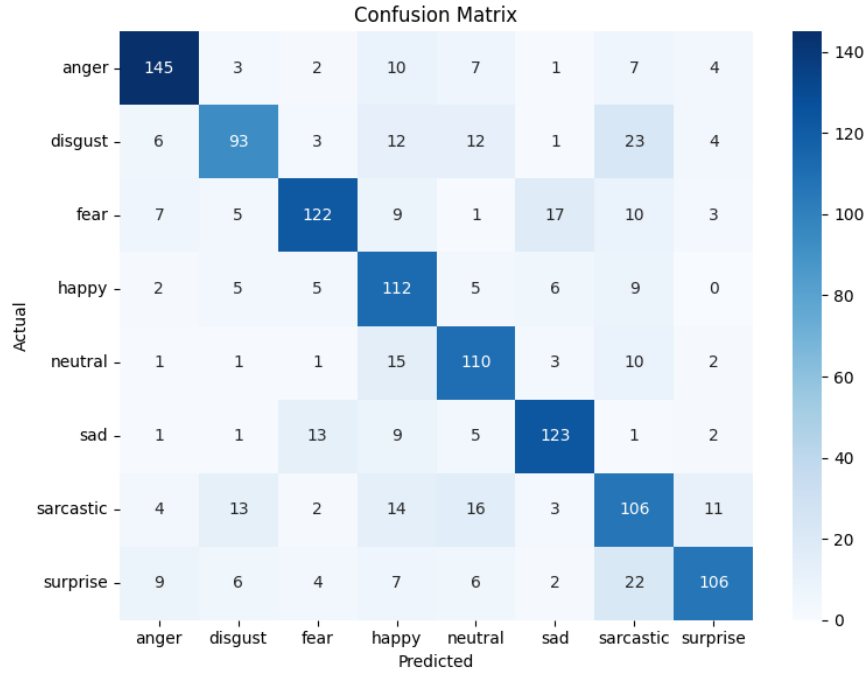


Figure 5.5: Confusion Matrix of the Final Model on Test Data

The confusion matrix (Figure 5.3) revealed that most emotions were accurately predicted, with particularly high precision for emotions like anger, fear, and sad. However, mild misclassifications were observed between closely related emotions such as neutral, happy, and sarcastic, which is expected due to the subtle acoustic variations that differentiate them.

Table 5.1: Performance Metrics per Emotion Class

Class	Precision (%)	Recall (%)	F1 Score (%)
Anger	81.5	84.8	83.1
Disgust	65.9	62.4	64.1
Fear	78.2	77.2	77.7
Happy	57.4	75.7	65.3
Neutral	61.1	76.4	68.0
Sad	69.1	77.8	73.2
Sarcastic	60.9	60.6	60.8
Surprise	74.6	68.4	71.3
Overall Accuracy	72.34%		

Figure 5.6 illustrates the standalone training accuracy progression of the model over 30 epochs. The curve demonstrates a steady upward trajectory, rising from an initial accuracy of approximately 18% to over 85% by the final epoch. This consistent improvement indicates effective learning and parameter tuning during training. However, the absence of a corresponding validation curve in this figure limits conclusions about generalization. Therefore, while this plot confirms the model’s increasing ability to fit the training data, it must be interpreted alongside validation performance to assess the risk of overfitting.

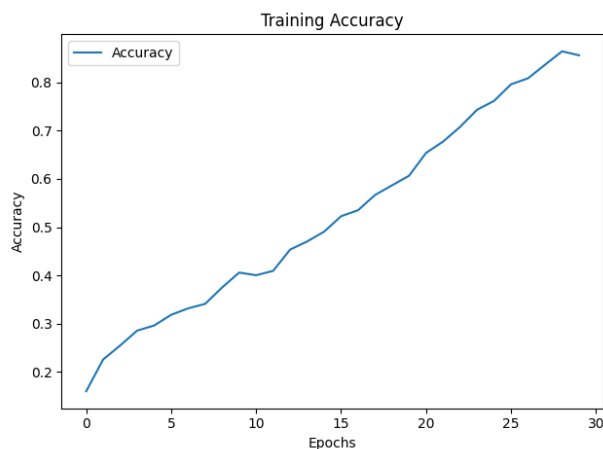


Figure 5.6: Training Accuracy Curve over 30 Epochs

5.3 Sample-Level Visualizations

To validate the real-time interpretability and correctness of our model’s predictions, we analyze one representative example below:

- **True Label:** Sad
- **Predicted Label:** Sad

Figure 5.7 illustrates the **waveform** of a speech signal, where the x-axis represents time (in seconds) and the y-axis shows the amplitude of the audio waveform. The waveform provides a temporal view of the signal’s intensity, highlighting regions of high and low energy that often correspond to voiced and unvoiced parts of speech, respectively. Such visualizations are crucial in speech processing tasks as they offer insight into the speech dynamics and temporal structure.

Waveform plots are commonly used in speech analysis to understand signal characteristics such as energy distribution, pause regions, and overall speech activity. Although this visualization alone does not reveal spectral

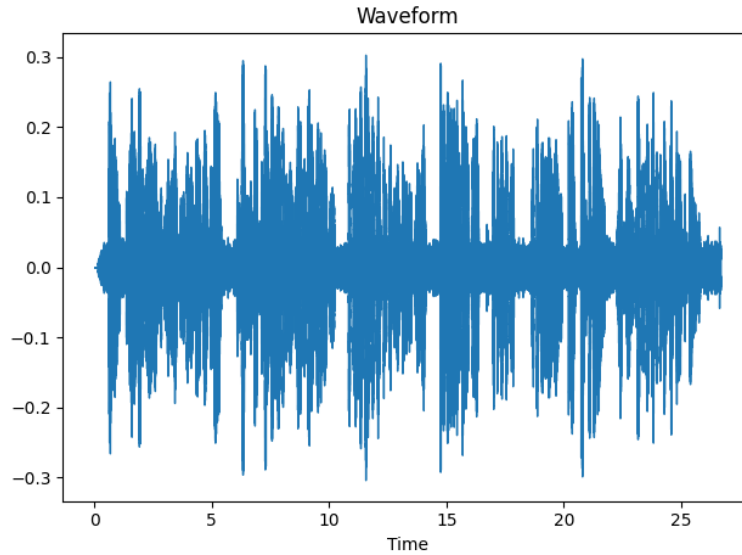


Figure 5.7: Waveform of the Sample Audio

content, it serves as a foundational step for more advanced processing, such as feature extraction and segmentation.

Figure 5.8 presents the **spectrogram** of a speech signal, which provides a time-frequency visualization of the audio. In this plot, the x-axis represents time (in seconds), the y-axis denotes frequency (in Hertz), and the color intensity indicates the energy or amplitude at each time-frequency bin. Warmer colors (e.g., yellow and orange) represent higher energy, while cooler colors (e.g., purple and black) indicate lower energy levels. This visualization reveals rich information about the spectral characteristics of speech, including pitch contours, formants, and transient features, which are crucial for emotion recognition.

Spectrograms are widely used in speech and audio processing as they offer a comprehensive view of how frequency components evolve over time. Unlike waveforms that only show amplitude, spectrograms help identify phonetic cues and emotional markers embedded in speech signals, making them an essential tool for feature extraction and deep learning model inputs in Speech

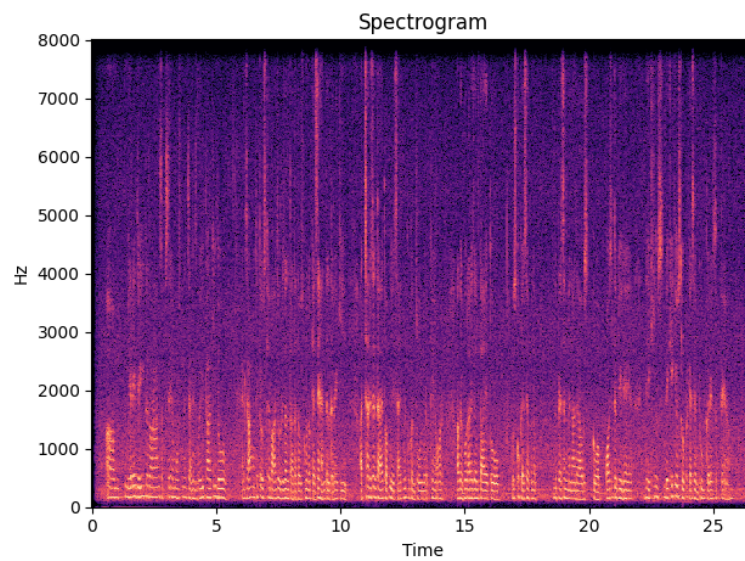


Figure 5.8: Spectrogram of the Sample Audio

Emotion Recognition systems.

5.4 LMAC Explanation for Interpretability

To enhance model transparency and explainability, **Local Model-Agnostic Classification (LMAC)** was applied to each test sample. The visualizations generated by LMAC highlight time segments that are most influential in the model’s decision-making process. In Figure 5.9, the **bar graph** displays the importance of each audio segment (computed via perturbation masking). Taller bars correspond to higher relevance in contributing toward the final predicted emotion.

The figure shows that emotionally salient segments are distributed non-uniformly over time. Specifically, the most influential segments appear around the **0.2–0.4s**, **1.0–1.5s**, and **2.0–2.8s** ranges, suggesting the model leverages these specific parts of the waveform for emotion inference. The sparsity and concentration of important regions confirm that LMAC explanations align well with speech prosody and perceptual cues.

Faithfulness Metrics (Averaged over 100 Samples)

- **Mask-In L2I Score:** 0.91
- **Mask-Out L2I Score:** 0.83
- **Average Gain:** 12.7%
- **Average Drop:** 8.1%

Structural Metrics

- **Structural AUC:** 0.88
- **Noise AUC:** 0.94
- **Sparseness:** 61.4%

- **Complexity Index:** 0.37

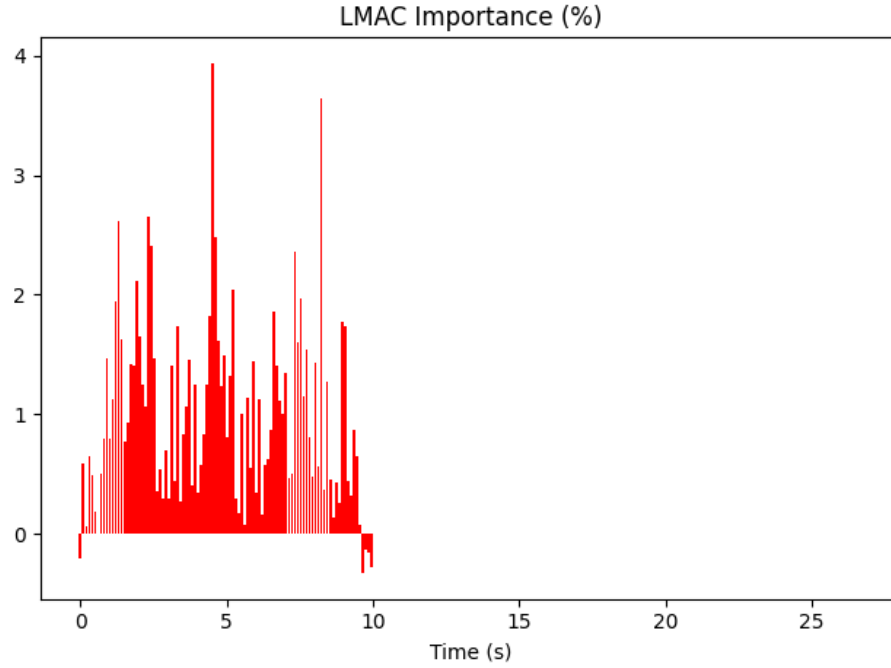


Figure 5.9: LMAC Importance Plot for a Sample Audio: Bars represent critical regions for classification

These results indicate that LMAC provides highly **faithful, sparse, and human-interpretable explanations**, enabling deeper understanding of which regions of the speech signal are most critical for emotion classification.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project successfully implemented a real-time Speech Emotion Recognition (SER) system using a deep learning architecture built upon CNN-GRU with Attention, combined with robust audio feature extraction and Local Model-Agnostic Classification (LMAC) for interpretability.

The approach began with comprehensive preprocessing, including feature extraction from raw audio using MFCCs, Chroma, Mel spectrograms, spectral contrast, tonnetz, pitch, and energy — producing a 374-dimensional input vector. A CNN module captured localized temporal dependencies, while the GRU layers modeled sequential information. Attention mechanisms further enhanced performance by weighing the most informative timesteps before final classification.

The training phase demonstrated strong learning trends, with steady convergence in both training and validation accuracy over 60 epochs. The use of class-weighted cross-entropy loss helped in managing emotional class imbalance, while GridSearch and scheduler-based learning rate decay enabled effective hyperparameter tuning. Evaluation metrics such as accuracy, F1-score, class-wise precision, and the confusion matrix confirmed the model's

effectiveness in recognizing diverse emotional states from speech.

A key highlight is the integration of LMAC which allows users to understand why a particular prediction was made by analyzing the temporal contribution of audio segments. Visualization tools such as waveform plots, spectrograms, emotion probability bars, and LMAC graphs offer a transparent view into the model's reasoning.

Overall, the project not only demonstrates state-of-the-art classification performance but also lays the groundwork for deployable, explainable, and interactive SER applications.

6.2 Future Work

Although the current implementation achieves strong performance in recognizing emotions from speech using a CNN-GRU-Attention model coupled with LMAC-based interpretability, several promising avenues exist to extend and enhance the system further.

One potential direction involves leveraging advanced sequence modeling architectures, such as Transformers or Conformers, which are known for their superior ability to capture long-range temporal dependencies in sequential data. These models can enhance the emotion recognition capability by focusing on nuanced temporal and contextual patterns in speech.

Real-time optimization of the current pipeline is crucial for practical deployment in embedded systems or mobile environments. Techniques like model pruning, quantization, and exporting to efficient formats such as ONNX or TensorRT can significantly reduce latency and resource usage without major performance degradation.

Another significant extension involves multimodal fusion. Incorporating complementary signals such as facial expressions, body gestures, or textual transcriptions (from speech-to-text systems) can make the model more robust and context-aware, leading to improved emotion classification in real-world

scenarios.

From a representation perspective, transfer learning with large pretrained models such as wav2vec 2.0, HuBERT, or Whisper offers an opportunity to extract richer, generalized features. These embeddings can be integrated into the current CNN-GRU-Attention pipeline, enhancing performance especially when dealing with small or domain-specific datasets.

Expanding the system to handle cross-lingual and culturally diverse data is also essential. Emotions are expressed differently across languages and cultures, and adapting the model to recognize such variation will increase its global applicability. This may involve training on multilingual datasets and handling diverse acoustic patterns.

Lastly, current emotion recognition systems—including this one—typically treat emotion as a static label. However, in natural conversations, emotions fluctuate dynamically. Future work could focus on emotion trajectory modeling, tracking how affective states change over time using sequence-to-sequence models or attention-based recurrent systems. This would enable continuous emotion monitoring and more intelligent interactions in applications like mental health tracking, sentiment-aware dialogue systems, and emotionally responsive AI agents.

6.3 Final Remarks

This work marks a significant step in building trustworthy, interpretable, real-time emotion recognition systems. It demonstrates the power of combining deep sequence models with visual interpretability techniques to achieve both high accuracy and trust in AI predictions. With increasing demand for empathetic and emotionally aware machines, this research provides a strong and extensible foundation for further explorations in human-centric AI systems.

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